

On the Irrelevance of Networks in Repeated Games*

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Abstract

When do interaction and information networks affect equilibrium outcomes in environments with many bilateral repeated games? Using a multimarket Cournot counterexample with symmetric firms and markets, I show that the ability of all pairs of opponents to sustain collusion in equilibrium depends on both which players interact and what each player observes about others' interactions. I then develop a general framework and identify a sufficient condition on stage-game payoffs—*Optimality of Nash Reversion*—under which network structure is irrelevant: an action profile is sustainable between all opponents if and only if it is sustainable in the corresponding two-player game.

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1 Introduction

Cooperation in repeated interactions depends not only on who interacts with whom, but also on the information players have about each other's past behavior. In many economic environments, both interactions and information flows are limited and local: firms, consumers, and agents are embedded in multiple relationships yet observe only fragments of others' actions. Whether cooperation can be sustained then depends jointly on who meets whom—the interaction network—and who learns about whose behavior—the information network. For example, a prominent firm may interact with several smaller, peripheral firms, while the smaller firms do not interact directly with each other. Similarly, when trying a new product or service, we often share our experience only with our friends rather than posting a public review. Based on this private communication, our friends decide whether to buy the product and may pass their impressions along to their friends, and so on. This paper asks a simple question: *When do these networks matter?*

A classic benchmark is provided by Bernheim and Whinston (1990), who study symmetric price competition under perfect monitoring. They show that when all markets and firms are identical, producing perfect substitutes in each market, collusion can be sustained in multiple markets if and only if it can be sustained in a single market—so multimarket contact is irrelevant for sustaining collusion. The intuition is simple: the gain from deviation and the loss from punishment both scale proportionally with the number of markets. Wolitzky (2021) notes that the same argument extends to any environment in which identical prisoner's dilemmas are played simultaneously within a population under perfect monitoring.

I first show that this stark result depends critically on a feature of their environment: each firm's minmax payoff is attained at a static Nash equilibrium of the stage game. I present a counterexample of collusive multimarket contact in Cournot competition with symmetric firms and markets. Each firm interacts independently with each other firm in a separate market. I show that for an intermediate range of discount factors, collusion cannot be sustained in an equilibrium of the single-market game, nor can it be sustained in the multimarket game in the absence of an information network. In contrast, with information linkages, firms can coordinate harsher punishments by pooling punishers' incentives to deviate across markets. As a result, collusion in all markets can be sustained in equilibrium.

The contrast between the Bernheim–Whinston environment and the Cournot counterexample clarifies when the network structure ceases to matter. The key lies in the relationship between minmax and Nash equilibrium payoffs in the stage game, as mentioned above. In price competition with homogeneous goods or in the standard prisoner's dilemma, each player's minmax payoff is a Nash equilibrium payoff of the stage game; this is not the case in Cournot competition. When the two coincide for each player—a property I term *Optimality of Nash Reversion*—the harshest credible

punishment within any bilateral relationship is simply to revert to a stage-game Nash equilibrium, which may depend on the identity of the deviator. Punishments are therefore static: they do not rely on intertemporal incentives nor on the pooling of incentives across opponents, and information networks cease to affect whether a given target profile is sustainable.

I formalize this insight in a general framework where players take on one of two roles: a row player or a column player. This distinction accommodates asymmetric stage games. Each row player plays an identical, bilateral repeated game independently with each player in some player-specific subset of column players defined by an undirected opponent network. There is a directed information network, and players can observe all of the past interactions of players in their out-neighborhoods of this network. There may be directed links between any pair of players, including players assigned to different roles. Under Optimality of Nash Reversion, I show that for any opponent and information networks, a target action profile between all pairs of opponents can be sustained in a perfect Bayesian equilibrium in pure strategies if and only if it can be sustained in the corresponding two-player repeated game.

The intuition for the main result is as follows. First, suppose that the target profile *can* be sustained in a subgame perfect equilibrium of the corresponding two-player game. Then we can construct an analogous equilibrium in the network game in which each player conditions behavior in each bilateral relationship solely on that relationship's own history. Second, suppose that the target profile *cannot* be sustained in a subgame perfect equilibrium of the two-player game. Under Optimality of Nash Reversion, in any equilibrium, the average punishment of a deviator across all her opponents can never be lower than the optimal bilateral punishment. When evaluating whether to deviate, a player may either defect simultaneously against all her opponents or only against a subset of them. Which deviation is most profitable is not essential to the argument: to rule out a strategy profile as an equilibrium, we simply need to show that a profitable deviation exists.² Under a simultaneous deviation, both cooperative and deviation payoffs scale linearly with the number of opponents, as in Bernheim and Whinston (1990). Hence, a profitable deviation in the two-player game implies there exists a profitable deviation in the network game.

Beyond its theoretical contribution, the analysis speaks to central questions in industrial organization concerning multimarket contact and tacit collusion. Empirical studies document that increased contact among firms operating in multiple markets often softens competition and raises prices, particularly in the airline industry (e.g., Evans and Kessides, 1994; Ciliberto and Williams, 2014; Remer and Orchinik, 2024). Bernheim and Whinston (1990) explain this by showing that when markets are asymmetric, multimarket contact allows for the pooling of incentives to deviate

²If a player is sufficiently patient, conditional on deviating, she may prefer to defect against only a strict subset of her opponents. This can occur even if non-victims punish her after detecting a deviation either directly or indirectly through the information network. The reason is that it may take multiple periods before the punishment propagates through the network, allowing her to continue receiving cooperative payoffs from some opponents in the meantime.

across markets on the equilibrium path, and can additionally enable more severe punishments when products are differentiated. I provide a complementary mechanism: when the market structure is such that each firm earns strictly positive profits in the static Nash equilibrium, information linkages can make the difference in whether collusion is sustainable. While Bernheim and Whinston (1990) demonstrate that harsher punishments are achievable under asymmetry, I show that the same is possible in fully symmetric settings. A notable feature of the airline industry is that firms typically set capacities before engaging in price competition, effectively competing in quantities (Kreps and Scheinkman, 1983). My Cournot counterexample therefore offers an alternative explanation for observed tacit collusion in markets that are not characterized by pure price competition.

The paper is organized as follows. Section 2 discusses related work on multimarket contact, community enforcement, and repeated games on networks. Section 3 presents a Cournot counterexample illustrating how interaction and information networks affect equilibrium outcomes. Section 4 then introduces the general model and key assumption on the stage-game payoffs. Section 5 combines an illustrative example with the main theorem, showing that when each player’s minmax payoff coincides with a static Nash equilibrium payoff, the ability to sustain cooperation is independent of each network. Finally, Section 6 concludes and outlines directions for future research. Additional proofs are collected in Appendix A.

2 Related Literature

Bernheim and Whinston (1990) show that with identical firms and markets under price competition, multimarket contact does not facilitate collusion; as Wolitzky (2021) notes, the same result extends to any environment in which identical prisoner’s dilemmas are played simultaneously under perfect monitoring—a special case of the sufficient condition identified here. Bernheim and Whinston (1990) further show that asymmetries in markets or firms can break this irrelevance.

There is also a large empirical literature examining how multimarket contact shapes competition across industries, particularly in airlines and other oligopolistic markets. In the airline industry, greater multimarket contact is typically associated with softer competition and higher fares (e.g., Evans and Kessides, 1994; Ciliberto and Williams, 2014; Remer and Orchinik, 2024). Similar patterns arise in banking and other local markets (Heggestad and Rhoades, 1978; Jans and Rosenbaum, 1997; Pilloff, 1999). By contrast, Mester (1987) finds that greater contact among U.S. banks is associated with more intense competition.³

More broadly, the paper relates to work on community enforcement and folk theorems under imperfect or local monitoring. A classic line of research shows how cooperation can be sustained in

³See also Bilotkach and Tabacco (2013), Balaguer and Ripollés (2021), Dranove, Gron and Mazzeo (2003), and Yan (2024) for additional evidence across sectors.

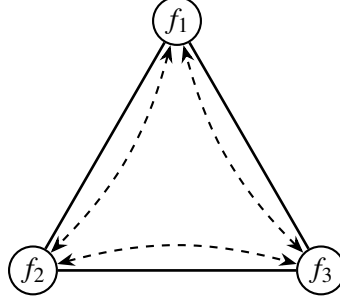
large populations through social norms or reputations when matching is anonymous and monitoring is indirect: for example, Kandori (1992) and Ellison (1994) study contagion-type strategies in random-matching environments, while Okuno-Fujiwara and Postlewaite (1995) embed reputational incentives in a status hierarchy. More recent contributions analyze folk theorems in settings where players observe only coarse or indirect signals of others' behavior. Deb, Sugaya and Wolitzky (2020) establish a folk theorem for prisoner's dilemma games under anonymous random matching with collective reputations, while Renault and Tomala (1998) and Laclau (2012, 2014) establish folk theorems when interaction and information structures are fixed via a network. Sugaya and Wolitzky (2020) show how the presence of a small fraction of behavioral types can sharply affect the scope for cooperation. These papers focus on characterizing equilibrium payoff sets for a given monitoring structure. By contrast, the main result here identifies a simple payoff condition—Optimality of Nash Reversion—under which who interacts with whom and who observes whom is irrelevant for sustaining a given outcome, and shows by counterexample how this irrelevance breaks down when the condition fails.

The paper also connects to models of repeated interaction on explicit networks, where the network determines who interacts with or observes whom. For example, Ahn and Suominen (2001), Lippert and Spagnolo (2011), Ali and Miller (2013, 2016), and Wolitzky (2013) study how local monitoring and network architecture affect cooperation, and Jackson and Zenou (2015) examine how network structure shapes efficient cooperation with local information. Most of this literature analyzes how network position and information flows affect equilibrium payoffs or contagion dynamics. The framework here instead disentangles the opponent network from the information network and establishes when this structure does not matter: under Optimality of Nash Reversion, each bilateral relationship is strategically independent, and the ability to sustain a given target profile reduces to that of the two-player game.

3 A Motivating Counterexample

There are three firms. In each period of an infinite horizon, each firm plays an identical symmetric discretized Cournot duopoly game separately with each of the other firms. Moreover, suppose that each firm observes all past quantities chosen in all three markets. Figure 1a illustrates these interaction and information structures: solid lines represent markets in which two firms meet, and dashed double arrows indicate that firms mutually observe each other's interactions. I first establish two preliminary claims about the single-market game and the multimarket game absent information linkages, before showing how information linkages facilitate collusion.

The underlying stage game is depicted in Figure 1b. In each market, each firm simultaneously chooses the quantity to produce of a homogeneous good, either low (L), medium (M), or high (H).



(a) Interaction and information structures. Each node f_i represents a firm. Solid lines indicate bilateral markets in which firms interact directly (the opponent network), while dashed double arrows indicate that firms mutually observe each other's behavior (the information network).

	L	M	H
L	12, 12	4, <u>14</u>	<u>0</u> , 11
M	<u>14</u> , 4	<u>7</u> , <u>7</u>	-4, 6
H	11, <u>0</u>	6, -4	-15, -15

(b) A symmetric discretized Cournot stage game. Static best-response payoffs are underlined. Unlike in prisoner's dilemma games, the unique Nash equilibrium (M, M) gives each firm strictly more than her minmax payoff. So Nash reversion is not the harshest credible punishment in the repeated game when firms are sufficiently patient, in either single- or multimarket settings.

Figure 1: The multimarket Cournot counterexample.

The minmax payoff is 0 for each firm, which is strictly lower than the unique static Nash equilibrium payoff of 7. I fix the common discount factor $\delta \in (1/7, 3/20)$. For this intermediate degree of impatience, the single-market repeated game admits no equilibrium outcome other than the static Nash equilibrium. In the multimarket game, the same discount factor is still too low to sustain the collusive outcome when the information network is absent, but it is high enough to sustain collusion once the information network is present. The proofs of all the following claims are in Appendix A.

Claim 1. *In the single-market game, the unique pure-strategy subgame perfect equilibrium is the stationary profile in which each firm plays M after every history.*

For this range of discount factors, Nash reversion is the harshest credible punishment available in the bilateral game, but it is not severe enough to deter deviation from any other profile. Hence, the unique pure-strategy subgame perfect equilibrium is that firms play the static Nash equilibrium after every history.

Consider the single-market game between f_1 and f_2 . To see why the claim holds, I first show that there does not exist a pure-strategy subgame perfect equilibrium in which f_1 's normalized discounted payoff is strictly less than her static Nash equilibrium payoff. Suppose instead that there

does. Then in the equilibrium yielding the lowest payoff to f_1 , (L, H) must be played in the first period. This is because to get a payoff strictly less than 7, the flow payoff of f_1 in her optimal punishment must be strictly less than 7.⁴ And for either (L, M) , (M, H) , or (H, H) , there would necessarily be a profitable one-shot deviation in this first period for at least one player.

Although there is no myopic incentive to deviate from (L, H) for f_1 , this incentive is present for f_2 . In order to deter a deviation of f_2 to M , we would need $\underline{u} < -3$. This comes from the inequality

$$\delta(14 - \underline{u}) \geq (1 - \delta)(14 - 11),$$

and setting $\delta < 3/20$. This constitutes a contradiction, since -3 is strictly less than each player's minmax payoff of 0.

Claim 2. *In the multimarket Cournot game without information linkages, there does not exist a pure-strategy perfect Bayesian equilibrium in which (L, L) is played in each period on the path of play by all pairs of opponents.*

Without information linkages, uninvolved firms can only learn of a deviation indirectly through the actions of others, necessarily introducing at least a one-period lag in their response. As a result, any credible punishment is insufficiently severe to deter deviation from the collusive profile.

Suppose instead that such an equilibrium exists. To deter a deviation by f_1 , following a deviation in a particular market, the other firm in that market must choose H in the first period of the punishment phase. If f_2 instead chose M or L against f_1 , then f_1 could first deviate against f_2 and, in the next period, deviate against f_3 while continuing to play the myopic best response to f_2 . Since f_3 does not directly observe the interaction between f_1 and f_2 , even if f_2 were able to signal f_1 's deviation through her action against f_3 , there would be a one-period lag before f_3 could punish f_1 . This deviation yields a payoff of at least

$$(1 - \delta)((14 + 12)/2 + \delta(14 + 7)/2) > 12,$$

for all $\delta \in (1/7, 3/20)$, contradicting the assumption that (L, L) is played on the equilibrium path. Moreover, in the first period of the punishment phase, f_1 must choose L against at least one opponent. Otherwise, following an initial deviation against both opponents at once, there would necessarily exist a profitable one-shot deviation for f_1 to L in both markets. But if (L, H) is played by f_1 and f_2 , then f_2 has a profitable one-shot deviation to M . This is because it would take at least one more period for f_1 to signal to f_3 via her action that f_2 deviated; in the meantime, f_2 can guarantee herself the same payoff against f_3 as if she had followed the prescribed profile.

⁴Otherwise, there would necessarily exist an equilibrium yielding a strictly lower payoff, contradicting the punitive optimality of the equilibrium under consideration.

Claim 3. *In the multimarket Cournot game with information linkages, there exists a pure-strategy subgame perfect equilibrium in which (L, L) is played in each period on the path of play by all pairs of opponents.*

With information linkages, all firms immediately observe any deviation, enabling a coordinated punishment response across all markets simultaneously. This makes the punishment sufficiently severe to deter deviation from the collusive profile.

Suppose that f_1 unilaterally deviates from (L, L) in at least one market in period t . Any such deviation is immediately observed by both f_2 and f_3 . Consider then the punishment sequence in which the pairs (f_1, f_2) and (f_1, f_3) immediately revert to the f_1 -minmax profile (L, H) forever onwards, while the pair (f_2, f_3) immediately reverts to the static Nash profile (M, M) forever onwards. If any player deviates from this path, an analogous punishment sequence is initiated for the new deviator. The average discounted payoff of f_1 induced by this punishment profile is exactly her minmax payoff of 0. The punishment is sufficient to deter a deviation from (L, L) in each market in each period, since

$$(1 - \delta)(14 - 12) < \delta(12 - 0)$$

for all $\delta \in (1/7, 3/20)$. It then remains to show that the punishment profile is indeed an equilibrium. Since L is a myopic best response to H , there is no profitable deviation for f_1 . And there is no profitable deviation for f_2 or f_3 either, since

$$(1 - \delta)((14 + 7)/2 - (11 + 7)/2) < \delta((11 + 7)/2 - 0)$$

for all $\delta \in (1/7, 3/20)$.

When players have multiple opponents, information linkages across games allow for the pooling of incentives to deter deviation from a costly punishment path. In each period of the punishment phase, each punisher has the myopic incentive to deviate in only one market at a time. And if a punisher deviates off the path of play, she is treated as the new deviator by all opponents who observe the deviation. So not only is she punished by the player she was supposed to be punishing, but also the other punisher.

In the next section, I introduce a general model of repeated bilateral interactions on a network. Guided by the contrast between the counterexample and the price competition setting of Bernheim and Whinston (1990), I provide a sufficient condition on the stage-game payoffs under which the network structure becomes irrelevant for sustaining a particular behavior in equilibrium: each player's minmax payoff must be a Nash equilibrium payoff of the stage game.

4 General Model

Let N denote the set of all players, where the set of row players is given by $R = \{r_1, \dots, r_n\}$, and the set of column players by $C = \{c_1, \dots, c_m\}$, such that $N = R \cup C$ and $R \cap C = \emptyset$. For $P = R, C$, let $-P = N \setminus P$.

Each row player simultaneously and independently plays a repeated bilateral stage game with a fixed subset of column players. Each player discounts the future according to the common discount factor $\delta \in (0, 1)$. The stage game is identical across pairs of opponents. Let A_R denote the set of row-player actions and A_C the set of column-player actions, which I assume to be finite. For $P = R, C$, the stage game payoff for each player $i \in P$ is given by the mapping $u_P : A_R \times A_C \rightarrow \mathbb{R}$. Moreover, the pure-action minmax payoff of player $i \in P$ is given by

$$\underline{v}_P \equiv \min_{a_{-P} \in A_{-P}} \max_{a_P \in A_P} u_P(a_P, a_{-P}).$$

The key assumption I make, which drives the main result, is that each player's minmax payoff is a Nash equilibrium payoff of the stage game.

Assumption 1 (Optimality of Nash Reversion). *For $P = R, C$, there exists $\underline{a}^P \in A_R \times A_C$ such that $\underline{a}^P$ is a Nash equilibrium of the stage game and $u_P(\underline{a}^P) = \underline{v}_P$.*

This assumption ensures that the harshest credible punishment that can be imposed within each bilateral relationship is Nash reversion. Hence, for every degree of impatience, optimal punishments do not rely on intertemporal incentives. This greatly simplifies the analysis, since, in general, the sequence of action profiles constituting an optimal punishment and its payoff are unknown for an arbitrary value of the discount factor, even if there are only two players. Moreover, it implies that for each player, the harshest average punishment a subset of opponents can credibly impose can never be worse for her than that which a single opponent can credibly impose.⁵

Pairs of opponents are given by an undirected *opponent network*, denoted by g_O . There exists a bilateral relationship between i and j if $(i, j) \in g_O$. A player $i \in P$ may only be linked to players in $-P$. The set of opponents of player i is then given by her *neighborhood* in g_O , denoted by $\mathcal{N}_{g_O}(i)$, where

$$\mathcal{N}_{g_O}(i) \equiv \{j \in -P : (i, j) \in g_O\}.$$

Her degree in the opponent network is denoted by

$$d_i \equiv |\mathcal{N}_{g_O}(i)|.$$

I assume that every player has at least one opponent: $d_i \geq 1$ for all $i \in N$.

⁵Indeed, in Section 3, I demonstrate that the latter may no longer hold when Assumption 1 is relaxed.

Remark 1. In the general model, I divide the set of players into row and column players to accommodate for both symmetric and asymmetric stage games. Restricting attention to symmetric stage games as in the Cournot counterexample of Section 3, it is straightforward that the arguments proving the main result in Section 5 hold if g_O allows links between any pair of distinct players in N .

In addition, players are related through a directed *information network*, denoted by g_I . For each player i , there may be a directed link from i to any player in $N \setminus \{i\}$. If $(i, j) \in g_I$, there is a directed link from player i to player j , and i perfectly observes all of player j 's past interactions. The set of players that $i \in N$ observes is given by her *out-neighborhood* in g_I , denoted by $\mathcal{N}_{g_I}^{\text{out}}(i)$, where

$$\mathcal{N}_{g_I}^{\text{out}}(i) \equiv \{j \in N \setminus \{i\} : (i, j) \in g_I\}.$$

Let a_{ij}^t denote player i 's action in the stage game against player j at time t , where $a_i^t = (a_{ij}^t)_{j \in \mathcal{N}_{g_O}(i)}$ is the vector of i 's actions against each of her opponents in period t , and $a^t = (a_i^t)_{i \in R \cup C}$ is the vector of all actions taken by each player in period t . For each pair of opponents $(k, j) \in g_O$, denote by

$$a_{(k,j)}^\tau \equiv (a_{kj}^\tau, a_{jk}^\tau)$$

the action profile played in the bilateral interaction between players k and j in period τ . Player i observes the set of links

$$L_i \equiv \{(k, j) \in g_O : k \in \mathcal{N}_{g_I}^{\text{out}}(i) \cup \{i\}\},$$

consisting of her own links together with those of her out-neighbors. Her private history at time t is then given by

$$h_i^t \equiv (a_\ell^\tau)_{\ell \in L_i, \tau < t}.$$

This consists of all past action profiles played within her own interactions as well as those of her out-neighbors prior to period t . Denote by H_i^t the set of all possible time- t private histories for player i , where $H_i = \bigcup_{t=1}^{\infty} H_i^t$ is the set of possible player- i histories across all periods.

I restrict attention to pure-strategy equilibria. A pure strategy for a player i is given by

$$\sigma_i : H_i \rightarrow A_P^{d_i}.$$

For each possible private history, player i chooses a pure action against each of her opponents. Let Σ_i denote the set of all pure strategies for player i . A strategy profile is denoted by $\sigma = (\sigma_i)_{i \in N}$. Let $\sigma_{ij}(h_i^t)$ be the component of $\sigma_i(h_i^t)$ corresponding to the action that player i takes against opponent j after private history h_i^t . And let $a_{ij}^t(\sigma)$ denote the period- t action of player i against player j induced by the strategy profile σ .

The *overall* period- t history is given by

$$h^t \equiv (a^\tau)_{\tau < t}.$$

This consists of all past actions of all players prior to t . Given the strategy profile σ , the continuation profile induced by h^t is denoted by $\sigma|_{h^t}$. Moreover, let $\mathbf{a}^t(\sigma)$ denote the overall history induced by σ .

I assume that a player's total payoff in a given period is additive in her bilateral stage-game payoffs; for convenience I normalize by her degree in the opponent network. So the average discounted payoff of a player $i \in P$ under the strategy profile σ is given by

$$U_i(\sigma) \equiv (1 - \delta) \sum_{t=1}^{\infty} \delta^{t-1} \left(\frac{1}{d_i} \sum_{j \in N_{gO}(i)} u_P(a_{ij}^t(\sigma), a_{ji}^t(\sigma)) \right).$$

Suppose that we are interested in supporting $\bar{a} = (\bar{a}_R, \bar{a}_C) \in A_R \times A_C$ by all pairs of opponents in a perfect Bayesian equilibrium, with

$$u_P(\bar{a}) =: \bar{u}_P > \underline{v}_P$$

for $P = R, C$.⁶ Moreover, the maximal myopic gain of each player $i \in P$ for deviating from $\bar{a}$ against a single opponent is denoted by

$$\gamma_P \equiv \max_{a_P \in A_P} u_P(a_P, \bar{a}_{-P}) - \bar{u}_P.$$

Remark 2. *In the corresponding two-player repeated game, there thus exists a subgame perfect equilibrium in which $\bar{a}$ is played in each period on the path of play if and only if $\delta \geq \max\{\underline{\delta}_R, \underline{\delta}_C\}$, where for $P = R, C$,*

$$\underline{\delta}_P \equiv \frac{\gamma_P}{\gamma_P + \bar{u}_P - \underline{v}_P}.$$

Having defined the general framework and key assumption, I now turn to the analysis. To make the logic of the result transparent, Section 5 begins with a simple illustration showing how, under role-specific Nash reversion, a profitable deviation against any subset of opponents implies a profitable simultaneous deviation against all opponents. The illustration serves to build intuition for the main theorem, which formally establishes that when each player's minmax payoff coincides with a static Nash equilibrium payoff, the ability to sustain a particular behavior in equilibrium is

⁶In the general network game, each player will not observe all past interactions. Perfect Bayesian equilibrium and subgame perfect equilibrium coincide when there are only two players or when the information network is sufficiently dense, as in the counterexample of Section 3.

independent of the structure of both networks.

5 Main Result: Network Irrelevance

This section states the main network irrelevance result. I first provide an example which illustrates the mechanics of the general model and the key intuition behind the result.

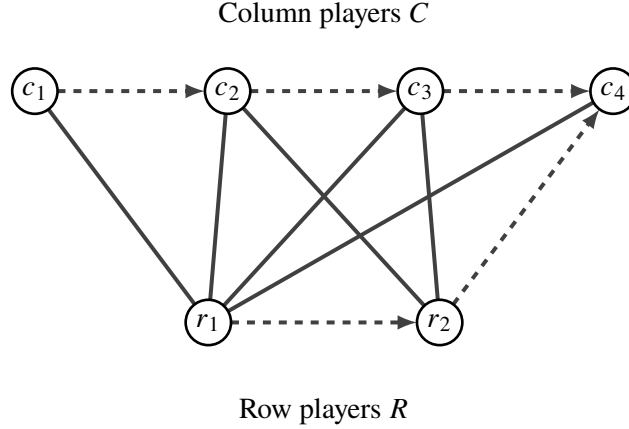
5.1 Illustration With Role-Specific Nash Reversion

Consider the 3×3 stage game in Figure 2b. The row player chooses between top (T), middle (M), and bottom (B), and the column player chooses between left (L), center (C), and right (R). Assumption 1 holds in a role-specific way: the game admits two pure-strategy Nash equilibria, (M, C) and (B, R) . (M, C) admits the column player's minmax payoff, whereas (B, R) admits that of the row player. So Nash reversion is an optimal bilateral punishment in the repeated game.

Set the target profile to $(T, L) \equiv \bar{a}$. Each player receives a payoff of 3, which is strictly greater than each player's minmax payoff of 1. The maximal myopic gain from deviating from this target profile is $4 - 3 = 1$ for the row player and $6 - 3 = 3$ for the column player. The row player's deviation is deterred for $\delta \geq 1/3$ and the column player's for $\delta \geq 3/5$. Therefore, the binding incentive constraint in the repeated two-player game is that of the column player: there exists a subgame perfect equilibrium in which (T, L) is played in each period on the path of play if and only if $\delta \geq 3/5$.

Now suppose this stage game is played repeatedly on the opponent network in Figure 2a. Two row players, r_1 and r_2 , each repeatedly interact with a player-specific subset of the column players, c_1, c_2, c_3 , and c_4 . The information network contains a directed line of the column players, such that each column player c_j directly observes the past play of her successor. Moreover, r_1 directly observes all past interactions of r_2 , and r_2 directly observes all past interactions of c_4 .

Suppose that r_1 considers whether to deviate from (T, L) against a subset of her opponents. Conditional on deviating, it is not immediate whether a *simultaneous* or a *sequential* deviation is more profitable: In a simultaneous deviation, r_1 deviates from $\bar{a}$ against all of her opponents in the same period, whereas under a sequential deviation, she deviates only against a strict subset of them. A simultaneous deviation by r_1 in the network game is equivalent to her deviation in the two-player game and is therefore deterred for $\delta \geq 1/3$. For $\delta > 1/3$, the relevant question becomes whether r_1 has a profitable sequential deviation, and which sequential deviation is most profitable can depend non-trivially on δ and on the information network. For example, suppose that r_1 begins by deviating against c_4 in period t . According to the information network, c_3, c_4 , and r_2 immediately detect the deviation. Note that r_2 can continue to cooperate with c_2 and c_3 following this deviation. In $t + 1$, the



(a) Opponent and information networks. Player r_1 interacts bilaterally with all c_j , whereas r_2 only interacts with c_2 and c_3 , indicated by solid lines. A dashed arrow from one player to another indicates that player directly observes all past interactions of the other, e.g., r_2 directly observes all past interactions of c_4 .

	L	C	R
T	3, 3	0, 0	0, <u>6</u>
M	<u>4</u> , 0	<u>2</u> , <u>1</u>	0, 0
B	0, 0	0, 0	<u>1</u> , <u>2</u>

(b) A 3×3 stage game with minmax-attaining Nash equilibria: (M, C) yields the column player's minmax payoff and (B, R) yields the row player's minmax payoff. Static best-response payoffs are underlined. Under Assumption 1, Nash reversion is an optimal punishment in the infinitely repeated two-player game.

Figure 2: The role-asymmetric illustration.

pairs (r_1, c_1) and (r_1, c_2) can continue to play (T, L) , whereas (r_1, c_3) and (r_1, c_4) can immediately revert to (B, R) . Player c_2 would then observe that c_3 is punishing r_1 , and can also punish her in $t + 2$.⁷ Similarly, player c_1 would observe that c_2 is punishing r_1 and follow suit in period $t + 3$. Instead of waiting for c_1 and c_2 to detect the deviation via other players' actions, suppose that r_1 preemptively plays M against both c_1 and c_2 in $t + 1$ while each of them continues to play L . The payoff from this sequential deviation is strictly more profitable than that of a simultaneous deviation for r_1 if and only if $\delta > 1/2$. Therefore, for $\delta \in (1/2, 1)$, the relevant incentive constraint for r_1 is necessarily given by the comparison between a sequential deviation payoff and the cooperative payoff.

However, we do not need to resolve this comparison directly. We already know from above that

⁷In the case that r_1 chooses T in $t + 1$ in an attempt to avoid being detected, we can specify a profile such that c_2 plays the punishment action R against r_1 whenever she observes that c_3 plays the punishment action against r_1 .

whenever $\delta \in (1/2, 3/5)$, there is a profitable simultaneous deviation for each of the column players, since their incentive constraint coincides with that of the column player in the two-player game. One of the key arguments of the main result relies on the fact that to disprove that a strategy profile is an equilibrium, it is sufficient to show that there exists at least one profitable deviation, regardless of what the most profitable deviation is for each player at a given discount factor. So we know that whenever we cannot sustain (T, L) in a subgame perfect equilibrium of the two-player game, we cannot sustain this profile between all pairs of opponents in a perfect Bayesian equilibrium of the network game.

I now present the formal result and describe the above intuition in more detail. I additionally provide intuition as to why the target profile can be sustained in the network game whenever it can be sustained in the corresponding two-player game.

5.2 The Main Theorem

Theorem 1 (Network Irrelevance). *Under Assumption 1 (Optimality of Nash Reversion), for any (N, g_0, g_1) , there exists a perfect Bayesian equilibrium in which $\bar{a}$ is played by all pairs of opponents in each period on the path of play if and only if such an equilibrium exists in the corresponding two-player game.*

To show necessity, note first that any credible collective punishment must give to each player at least her minmax payoff, which follows immediately from the definition of the minmax payoff. By Assumption 1, for any degree of impatience, the optimal punishment in the two-player game is reversion to the minmax-inducing Nash equilibrium of the deviator. It follows that the average collective punishment following a deviation in the network game can never be worse for the deviator than that of the two-player game. Therefore, if there exists an equilibrium in which the target profile can be sustained in the network game, the discount factor must be at least as high as the critical threshold above which it can be sustained in the two-player game.

To show sufficiency, note that when the discount factor is above the critical threshold of the two-player game, we can always construct a perfect Bayesian equilibrium where within each bilateral interaction, future behavior only depends on past play within that same interaction. By construction, the opponent and information networks become strategically irrelevant, and hence we recover the same incentive constraints as in the two-player game.

This result implies that information linkages are irrelevant whenever Nash reversion is always an optimal bilateral punishment. Even with local or asymmetric monitoring, the ability to sustain cooperation depends solely on the bilateral incentives of each pair. Indeed, the central insight from the irrelevance result of Bernheim and Whinston (1990) goes through: although it is possible to impose harsher absolute punishments on deviators, if a player deviates against all opponents simultaneously,

the average gain and punishment from deviating are identical to those of the two-player case. However, when the key assumption does not hold, as demonstrated in the counterexample of Section 3, optimal punishments in the network game generally rely on intertemporal coordination across opponents. In this sense, Assumption 1 delineates when information linkages can matter: when it holds, they never do; when it fails, the counterexample shows that they can.

Proof. Necessity. Let σ^* be a strategy profile such that for $P = R, C$, for each $i \in P$, $j \in \mathcal{N}_{go}(i)$, and $t \in \mathbb{N}$,

$$a_{ij}^t(\sigma^*) = \bar{a}_P.$$

Let

$$\tilde{\sigma}_i \in \arg \max_{\sigma_i \in \Sigma_i} \left\{ U_i(\sigma_i, \sigma_{-i}^*) : a_{ij}^1(\sigma_i, \sigma_{-i}^*) \in \arg \max_{a_P \in A_P} u_P(a_P, \bar{a}_{-P}) \quad \forall j \in \mathcal{N}_{go}(i) \right\}. \quad (1)$$

The maximum is attained since actions are finite and payoffs are discounted, so $\tilde{\sigma}_i$ is well-defined; it is the most profitable deviation for i amongst those that myopically best-respond to $\bar{a}_{-P}$ against every opponent in the first period. Suppose that σ^* is a PBE profile. Then

$$\begin{aligned} U_i(\sigma^*) &\geq U_i(\tilde{\sigma}_i, \sigma_{-i}^*) \\ \iff \bar{u}_P &\geq (1 - \delta)(\bar{u}_P + \gamma_P) + \delta U_i\left((\tilde{\sigma}_i, \sigma_{-i}^*)|_{\mathbf{a}^2(\tilde{\sigma}_i, \sigma_{-i}^*)}\right) \\ \iff \delta &\geq \frac{\gamma_P}{\bar{u}_P + \gamma_P - U_i\left((\tilde{\sigma}_i, \sigma_{-i}^*)|_{\mathbf{a}^2(\tilde{\sigma}_i, \sigma_{-i}^*)}\right)} \\ \implies \delta &\geq \frac{\gamma_P}{\bar{u}_P + \gamma_P - \underline{v}_P} \equiv \underline{\delta}_P. \end{aligned}$$

The last inequality follows from $U_i\left((\tilde{\sigma}_i, \sigma_{-i}^*)|_{\mathbf{a}^2(\tilde{\sigma}_i, \sigma_{-i}^*)}\right) \geq \underline{v}_P$. (Otherwise, (1) would be violated by the definition of the minmax payoff $\underline{v}_P$.) Therefore, the set of equilibria in which $\bar{a}$ is played by all pairs of opponents in each period on the path of play is non-empty only if $\delta \geq \max\{\underline{\delta}_R, \underline{\delta}_C\}$.

Sufficiency. Suppose the last inequality above holds for both roles $P = R, C$. For each pair of opponents $(i, j) \in g_O$ with $i \in P$, and each history, define the *first unilateral deviation date* in the interaction between i and j as

$$D_{ij}^t \equiv \min \left\{ \tau < t : \text{exactly one of } a_{ij}^\tau \neq \bar{a}_P \text{ and } a_{ji}^\tau \neq \bar{a}_{-P} \text{ holds} \right\},$$

with $D_{ij}^t = \infty$ if no such period exists. Note that D_{ij}^t does not vary with t once it is finite, so the identity of the date- D_{ij}^t deviator is well defined. Construct the *victims-only grim trigger* strategy $\bar{\sigma}$

such that for $P = R, C$, for each $i \in P$, $j \in \mathcal{N}_{go}(i)$, and $h_i^t \in H_i$,

$$\bar{\sigma}_{ij}(h_i^t) = \begin{cases} \underline{a}_P^P & \text{if } D_{ij}^t < \infty \text{ and the date-} D_{ij}^t \text{ deviator is } i, \\ \underline{a}_P^{-P} & \text{if } D_{ij}^t < \infty \text{ and the date-} D_{ij}^t \text{ deviator is } j, \\ \bar{a}_P & \text{otherwise.} \end{cases}$$

On the path of play, every pair of opponents plays $\bar{a}$. Within each bilateral interaction, play switches permanently to the static Nash equilibrium profile $\underline{a}^P$ following the first *unilateral* deviation in that interaction when the deviator is $i \in P$, and to $\underline{a}^{-P}$ when the deviator is $j \in -P$. Joint deviations, in which both members of a pair deviate in the same period, leave the prescription unchanged: the pair continues to play $\bar{a}$, and the trigger applies to the first subsequent unilateral deviation. Since $\bar{\sigma}_{ij}$ conditions only on $\left(a_{(i,j)}^\tau\right)_{\tau < t}$, which is a component of h_i^t for every L_i , the information network g_I is strategically irrelevant under this profile.

For player i , a deviation against any $k \leq d_i$ links yields an average discounted gain of $(1 - \delta) \frac{k}{d_i} \gamma_P$, and an average discounted loss of $\delta \frac{k}{d_i} (\bar{u}_P - \underline{v}_P)$. The factor $\frac{k}{d_i}$ cancels, so the incentive constraint is satisfied by construction. Histories at which $D_{ij}^t < t$ prescribe an absorbing repetition of static Nash equilibrium on that link, so play there is trivially sequentially rational. At every other history, including those reached through joint deviations, the continuation on each link coincides with the on-path profile, so the incentive constraint above applies unchanged. Finally, since payoffs are additively separable across links and continuation play on each link depends only on that link's own history, the one-shot deviation principle applies link by link, and checking single-period deviations against subsets of opponents is without loss. The profile $\bar{\sigma}$ is thus sequentially rational for every history, and constitutes a perfect Bayesian equilibrium profile sustaining $\bar{a}$ on all opponent links. $\square$

6 Conclusion

Opponent and information networks play distinct but complementary roles in shaping cooperation within a population. The opponent network determines who can affect whose payoffs, and hence which punishments are feasible. The information network determines who learns about whose deviations, and hence which punishments are credible. Taken together, these networks create two opposing forces. On the one hand, having more opponents and richer observability makes it possible to construct more severe punishments by coordinating behavior across multiple relationships. On the other hand, conditional on deviating, harsher punishments and a larger set of opponents make simultaneous deviation against all opponents more attractive than deviating in a single relationship.

This paper shows that how these forces balance depends crucially on the payoff structure of the underlying stage game. When optimal punishments necessarily rely on intertemporal coordination across opponents, the combined interaction–information structure can strictly expand the set of equilibrium outcomes. When optimal punishments are static, these forces exactly offset, and network structure becomes irrelevant for sustaining a particular behavior by all pairs of opponents. The Cournot counterexample illustrates the first case: with symmetric firms and markets, multimarket contact together with shared observability allows firms to pool incentives in the punishment phase and impose harsher average punishments than in any single market. In contrast, the general irrelevance result identifies a sufficient condition on the stage-game payoffs—Optimality of Nash Reversion, where each player’s minmax payoff coincides with a stage-game Nash equilibrium payoff—under which network structure does not matter for sustaining a particular population-wide behavior. When this condition holds, the harshest credible bilateral punishment is static Nash reversion for any degree of impatience, so each relationship is strategically independent and an action profile can be sustained between all pairs of opponents if and only if it can be sustained in the corresponding two-player repeated game.

These findings speak to central questions in industrial organization concerning multimarket contact and tacit collusion. In symmetric price competition, the minmax payoff of each firm coincides with its static Nash equilibrium payoff, so Optimality of Nash Reversion holds. Additional markets and richer observability may change which firms participate in punishing a deviator, but they cannot reduce her average continuation payoff below the bilateral Nash-reversion level. As a result, expanding the opponent network through multimarket contact or enriching the information network through greater observability has no independent effect on collusive stability, consistent with the benchmark irrelevance result of Bernheim and Whinston (1990). In the Cournot environment, by contrast, minmax payoffs lie strictly below static Nash equilibrium payoffs, so punishments that exploit coordination across opponents can be strictly harsher on average than any bilateral Nash-reversion punishment. Multimarket contact and information linkages then allow firms to pool incentives in the punishment phase and implement these harsher punishments, facilitating collusion even under symmetry. This mechanism helps explain why multimarket contact may or may not soften competition across industries, and suggests that the scope for coordinated effects following mergers depends not only on changes in market overlap but also on whether the underlying market structure permits harsher average punishments via coordination across relationships.

A natural next step is to characterize how alternative information networks affect equilibrium outcomes for a fixed interaction structure. The analysis here is fully general in strategies; in practice, however, a useful first step is to fix a stage game, discount factor, and opponent network, and then compare different information topologies within natural classes of strategies, such as simple penal codes that revert to a fixed action profile, as in the Cournot example. Such comparisons would

help isolate which features of the information network—such as density, the presence of highly informed players, or redundancy of monitoring paths—are most important for expanding the set of equilibrium outcomes.

A Proofs

A.1 Proof of Claim 1

For each $a \in \{L, M, H\}^2$, denote by $u_i(a)$ the stage-game payoff of firm i . Thus, the average discounted payoff of f_i induced by the strategy profile σ is given by

$$U_i(\sigma) \equiv (1 - \delta) \sum_{t=1}^{\infty} \delta^{t-1} u_i(a^t(\sigma)),$$

where $a^t(\sigma)$ is the period- t action profile induced by σ . The period- t history induced by σ is given by

$$\mathbf{a}^t(\sigma) \equiv (a^1(\sigma), \dots, a^{t-1}(\sigma)).$$

Given σ , let $\sigma|_h$ denote the induced continuation strategy following history $h \in H$. Let $\underline{\sigma} = (\underline{\sigma}^1, \underline{\sigma}^2)$ be an optimal penal code of the single-market game, where

$$\underline{u} := U_1(\underline{\sigma}^1) \equiv U_2(\underline{\sigma}^2).$$

The equivalence holds by the symmetry of the stage game. Denote by $\mathcal{F}^*$ the set of feasible, individually rational payoffs.

First, suppose there exists a SPE $\sigma^{H,H}$ such that $a^1(\sigma^{H,H}) = (H, H)$. This implies there exists $(w_1, w_2) \in \mathcal{F}^*$ such that

$$(1 - \delta)(0 - (-15)) \leq \delta(w_1 - 0),$$

so $w_1 \geq 15(1 - \delta)/\delta$. For all $\delta \in (1/7, 3/20)$, this requires $w_1 > 14$, contradicting feasibility. Hence (H, H) is never played in any SPE.

Second, suppose there exists a SPE $\sigma^{M,H}$ such that $a^1(\sigma^{M,H}) = (M, H)$. This implies that there exists $(w_1, w_2) \in \mathcal{F}^*$ such that

$$(1 - \delta)(0 - (-4)) \leq \delta(w_1 - 0),$$

so $w_1 \geq 4(1 - \delta)/\delta$. Again, it is clear that for all $\delta < 3/20$, this requires $w_1 > 14$, contradicting feasibility. So (M, H) is never played in any SPE. By symmetry, (H, M) is never played in any SPE either.

Now, I show that $\underline{u} = 7$. First, suppose instead that $\underline{u} < 7$. This implies that $a^1(\underline{\sigma}^1) \in \{(L, M), (L, H)\}$. (If $a^1(\underline{\sigma}^1) = (M, M)$, then $U_1(\underline{\sigma}^1|_{\mathbf{a}^2(\underline{\sigma}^1)}) < \underline{u}$, contradicting that $\underline{\sigma}$ is an optimal penal code. And clearly, if $a^1(\underline{\sigma}^1) \in \{(L, L), (M, L), (H, L)\}$, $U_1(\underline{\sigma}^1) > 7$, a contradiction.)

If $a^1(\underline{\sigma}^1) = (L, M)$, there is no profitable one-shot deviation by f_1 in period 1 if and only if

$$\begin{aligned} (1 - \delta)(7 - 4) &\leq \delta \left(U_1(\underline{\sigma}^1 | \mathbf{a}^2(\underline{\sigma}^1)) - \underline{u} \right) \\ \iff U_1(\underline{\sigma}^1 | \mathbf{a}^2(\underline{\sigma}^1)) &\geq 3(1 - \delta)/\delta + \underline{u}. \end{aligned} \quad (2)$$

Since $\forall (w_1, w_2) \in \mathcal{F}^*$, $w_1 \leq 14$, (2) and $\delta \in (1/7, 3/20)$ imply

$$\underline{u} \leq -3.$$

This constitutes a contradiction since -3 is less than the minmax payoff of 0.

So it must be that $a^1(\underline{\sigma}^1) = (L, H)$. Then there is no profitable one-shot deviation for firm f_2 if and only if

$$\begin{aligned} (1 - \delta)(14 - 11) &\leq \delta \left(U_2(\underline{\sigma}^1 | \mathbf{a}^2(\underline{\sigma}^1)) - \underline{u} \right) \\ \iff U_2(\underline{\sigma}^1 | \mathbf{a}^2(\underline{\sigma}^1)) &\geq 3(1 - \delta)/\delta + \underline{u}. \end{aligned} \quad (3)$$

Since $\forall (w_1, w_2) \in \mathcal{F}^*$, $w_2 \leq 14$, (2) and $\delta \in (1/7, 3/20)$ imply

$$\underline{u} \leq -3,$$

a contradiction.

So it must be that $\underline{u} \geq 7$. Clearly, there exists a SPE σ^1 such that $U_1(\sigma^1) = 7$: (M, M) is a Nash equilibrium of the stage game and $u_1(M, M) = 7$.

It is clear from the above argument that neither (L, H) nor (L, M) are ever played in any SPE. And by symmetry, neither (H, L) nor (M, L) are ever played in any SPE either.

Finally, suppose there exists a SPE $\sigma^{L,L}$ such that $a^1(\sigma^{L,L}) = (L, L)$. The one-shot deviation constraint for f_1 in period 1 requires

$$(1 - \delta)(14 - 12) \leq \delta \left(U_1(\sigma^{L,L} | \mathbf{a}^2(\sigma^{L,L})) - \underline{u} \right),$$

so the continuation payoff difference must satisfy

$$U_1(\sigma^{L,L} | \mathbf{a}^2(\sigma^{L,L})) - \underline{u} \geq 2 \frac{1 - \delta}{\delta}.$$

For all $\delta \in (1/7, 3/20)$, the right-hand side exceeds 11, while $U_1(\cdot) \leq 14$ and $\underline{u} = 7$ imply the left-hand side is at most 7, a contradiction. Hence (L, L) is never played in any SPE.

Therefore, the unique pure-strategy SPE σ^* is given by $\sigma_i(h) = M$ for all $h \in H$ for $i = 1, 2$,

yielding payoffs $U_1(\sigma^*) = U_2(\sigma^*) = 7$.

A.2 Proof of Claim 2

Suppose instead that there exists a perfect Bayesian equilibrium such that (L, L) is played in each period on the path of play by all pairs of opponents. Denote by σ^* an equilibrium strategy profile satisfying this condition.

Consider the decision of f_1 whether to deviate on the path of play. First, suppose that σ^* specifies that either L or M be played by f_2 in the period immediately following a unilateral deviation by f_1 against f_2 . Since f_3 does not directly observe the deviation by f_1 against f_2 , and since (L, L) is specified on the path of play, she must continue to play L against f_1 . So in the period immediately following the deviation, f_1 can get a payoff of 14 from f_3 by choosing M in that market. If σ^* specifies either L or M by f_2 against f_1 following the deviation, then f_1 can guarantee a payoff in the period immediately following the deviation of at least 7 against f_2 by choosing M in that market. Then by choosing M first only against f_2 and then again in the next period against both f_2 and f_3 , f_1 can guarantee herself from the initial deviation period onwards an average discounted payoff of at least

$$(17/20)\left((14 + 12)/2 + (3/20)(7 + 14)/2\right) \approx 12.39 > 12,$$

a contradiction. So σ^* must specify H be played by f_2 in the period immediately following a unilateral deviation by f_1 against f_2 on the path of play. By symmetry, H must also be played by f_3 in the period immediately following a unilateral deviation by f_1 against f_3 on the path of play under σ^* .

Moreover, suppose that σ^* specifies that in the period immediately following a deviation against both opponents, L is played against neither of them. Then there is necessarily a profitable one-shot deviation in the first period of the punishment phase: The average discounted gain from deviating to L against both opponents in the first period of the punishment phase would be at least $(17/20)(0 - (-4)) = 17/5$, whereas the loss cannot be more than $(3/20)(14 - 0) = 21/10$ according to the set of feasible payoffs.

In the period immediately following a unilateral deviation by f_1 against both f_2 and f_3 , suppose that under σ^* , f_1 and f_2 play (L, H) . But then, f_2 has the incentive to deviate to M in the first period of the punishment phase against f_1 : Since f_3 does not directly observe the interaction between f_1 and f_2 , f_2 can guarantee herself the payoff she would have received in her interaction with f_3 for at least two more periods. Denote by w_1 and w_2 the payoff of f_2 against f_3 in the first and second periods of the punishment phase respectively. By deviating to M in the first period of the punishment phase against f_1 , f_2 guarantees herself from the first period of the punishment phase

onwards an average discounted payoff of at least

$$(1 - \delta) \left((14 + w_1)/2 + \delta(w_2/2) \right).$$

By following the punishment f_2 gets an average discounted payoff of at most

$$(1 - \delta) \left((11 + w_1)/2 + \delta(14 + w_2)/2 \right) + \delta^2(14).$$

There thus exists a profitable deviation for f_2 if

$$(1 - \delta) \left((14 + w_1)/2 + \delta(0 + w_2)/2 \right) > (1 - \delta) \left((11 + w_1)/2 + \delta(14 + w_2)/2 \right) + \delta^2(14).$$

One obtains after rearranging

$$14\delta^2 + 17\delta - 3 < 0.$$

The positive root of the corresponding quadratic equation is $\delta^* = (\sqrt{457} - 17)/28 \approx 0.15634$. So the inequality holds for all $\delta \in (0, \delta^*)$. Since $\delta \in (1/7, 3/20)$ lies in this interval, a profitable deviation for f_2 exists for all discount factors under consideration. This constitutes a contradiction, proving the claim.

A.3 Proof of Claim 3

Consider the simple penal code in the sense of Abreu (1988), $\sigma = (\sigma^1, \sigma^2, \sigma^3)$, defined by the profile of punishment paths (Q^1, Q^2, Q^3) . For $i = 1, 2, 3$, let $Q^i = (q^i(t))_{t=1}^\infty$, where $q^i(t) = ((q_{12}^i(t), q_{21}^i(t)), (q_{13}^i(t), q_{31}^i(t)), (q_{23}^i(t), q_{32}^i(t)))$ is the period- t action profile induced by σ^i and $q_{jk}^i(t)$ the induced period- t action of firm j against firm k . Suppose that Q^1 , Q^2 , and Q^3 are as follows:

$$\begin{aligned} Q^1 &= \left(((L, H), (L, H), (M, M)), \dots \right), \\ Q^2 &= \left(((H, L), (M, M), (L, H)), \dots \right), \\ Q^3 &= \left(((M, M), (H, L), (H, L)), \dots \right). \end{aligned}$$

Let $u_i(a)$ denote the stage-game payoff of firm i given action profile $a \in \{L, M, H\}^2$ as defined in the payoff matrix of Figure 1b. Moreover, let

$$U_i(\sigma) = (1 - \delta) \sum_{t=1}^{\infty} \delta^{t-1} \left(\frac{1}{2} \sum_{j \in \{1, 2, 3\} \setminus \{i\}} u_i(a_{ij}^t(\sigma), a_{ji}^t(\sigma)) \right)$$

be the average discounted payoff of firm i induced by strategy profile σ , where $a_{k\ell}^t(\sigma)$ is the period- t action of firm k against firm ℓ induced by σ . It is easy to check that for $i = 1, 2, 3$, $U_i(\underline{\sigma}^i) = 0$.

Moreover, let σ^* be a strategy profile such that (L, L) is played by each pair of opponents on the path of play and such that $\underline{\sigma}^i$ is played following a unilateral deviation by i . We thus have $U_i(\sigma^*) = 12$.

On the path of play, the most profitable deviation for each firm is to choose M in both markets, yielding an average myopic gain of 2. However, in the following period, firm i gets an average discounted payoff of 0 rather than 12. There is no profitable deviation by any firm from σ^* on the path of play if and only if

$$(1 - \delta)(2) \leq \delta(12 - 0) \iff \delta \geq 1/7,$$

which is satisfied for all $\delta \in (1/7, 3/20)$.

It remains to check that for $i = 1, 2, 3$, there is no profitable deviation from $\underline{\sigma}^i$. Suppose that firm i unilaterally deviates from σ^* in period t . Then under σ^* , the firms begin playing according to $\underline{\sigma}^i$ in period $t + 1$. According to $\underline{\sigma}^i$, following a unilateral deviation by any firm k , firms begin playing according to $\underline{\sigma}^k$ in the following period, yielding to the subsequent deviator an average discounted payoff of $U_k(\underline{\sigma}^k)$ in the following period. Since L is a myopic best response to H , there is no profitable deviation for firm i . For each firm $k \neq i$, there is no profitable deviation from $\underline{\sigma}^i$ if and only if

$$(1 - \delta)(3/2) \leq \delta((11 + 7)/2) \iff \delta \geq 1/7,$$

which is satisfied for all $\delta \in (1/7, 3/20)$.

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